AMAN KUMAR

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EDUCATION

Board/Degree	Year	Name of Institution	Score
B.Tech (Information Technology)	2023-2027	Guru Gobind Singh Indraprastha University, New Delhi	CGPA – 9.0
C.B.S.E. (Class XII)	2022	Notre Dame School, Delhi	Percentage – 90.2%
C.B.S.E. (Class X)	2020	Notre Dame School, Delhi	Percentage – 94%

EXPERIENCE

Defence Research and Development Organisation (DRDO) – Software Development Intern, New Delhi (Jun 2025 – Jul 2025)

- Worked on a **Host-Based Intrusion Detection System (HIDS)** using **OSSEC**.
- Performed in-depth log analysis and debugging, reducing **false positives** by **~20%** and improving detection accuracy.
- Developed custom **OSSEC** rules, increasing detection coverage for suspicious activities by **~15%**.
- Worked in a Linux environment using shell commands and log files for troubleshooting.

PROJECTS

CODEBOARD – Developer Portfolio & Global Leaderboard – React.js, Node.js, MongoDB, Recharts, Axios, Node-cron, Postman Live

- Built a **MERN-based** platform aggregating coding stats for **500+ students** across LeetCode, HackerRank, and GeeksForGeeks.
- Integrated APIs and custom scrapers to sync real-time problem-solving and contest data.
- Tested and validated REST API endpoints and scraper responses with **Postman**, catching malformed and inconsistent data from third-party platforms before it reached the leaderboard.
- Developed a **C Score ranking algorithm** based on problem volume, ratings, and consistency.
- Implemented automated data refresh **every 6 hours** using **node-cron**.
- Designed an interactive dashboard and **real-time leaderboard** with multi-parameter filtering.
- Added a **Gemini AI-powered coding coach** that analyzes users' current stats and generates tailored practice recommendations.

ASK YOUR NOTES – Retrieval-Augmented Generation (RAG) System – Python, Streamlit, FAISS, Google Gemini API, Sentence-Transformers, PyPDF Github

- Engineered an end-to-end **RAG** application in **pure Python** to query lecture PDFs, giving full control over the pipeline and verifiable **[file, p.X] citations** on every answer.
- Implemented **sliding-window chunking** (500 chars, 80-char overlap) and 384-dim embeddings with **all-MiniLM-L6-v2**, preserving semantic continuity across page boundaries.
- Built a **FAISS** (IndexFlatL2) retrieval engine on L2-normalized vectors for exact cosine-similarity search over document chunks.
- Enforced strict **prompt guardrails** and **prompt-injection defenses** with the Gemini API, restricting answers to retrieved context and refusing out-of-scope queries instead of hallucinating.
- Designed a **Streamlit UI** with an excerpt inspector showing retrieved passages and similarity scores; benchmarked on a ground-truth set, achieving **100% Hit Rate @ 4** and **1.0 MRR**.

WALMART SALES ANALYSIS – SQL Data Exploration & Insights – SQL Github

- Analysed Walmart sales data across 3 branches (Mandalay, Yangon, Naypyitaw) with **1,000+ transactions** and 17 attributes.
- Performed **feature engineering** to extract time_of_day, day_name, and month_name columns for deeper trend analysis.
- Wrote **30+ optimised SQL queries** for product performance, revenue trends, customer segmentation, and VAT analysis.
- Conducted revenue and profit calculations including COGS, VAT, **gross margin** percentage, and identified top-performing product lines and peak sales periods.

SKILLS

Core Courses: Data Structures and Algorithms, Object Oriented Programming, Database Management Systems, Computer Networks
Extra Courses: Software Engineering
Languages: C++, JavaScript, Python
Frameworks and Libraries: Node.js, Express.js, React.js, MongoDB, MySQL, HTML, CSS, REST API
Generative AI: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Agentic AI
Cloud & Tools: AWS, Git, GitHub, Jira, Postman, Windows
Certifications: AWS Certified Cloud Practitioner Essentials, SQL Intermediate

ACHIEVEMENTS

ACADEMIC:

- Qualified **GATE 2026** (Computer Science).
- Secured **All India Rank 35** as a team in ZS Campus Beats Challenge.
- Participated in **10+ hackathons** across national and college levels.
- Maintained a strong academic record by maintaining a **CGPA of 9** (up to 6th semester), ranking consistently among **Top 15** students of the Information Technology department.
- National Service Scheme, BPIT – Responsible for society operations.

PROGRAMMING:

- Solved **500+ problems** on LeetCode with a rating of **1,826** and active on CodeChef with a rating of **1,472**.
- Contributed to open-source projects during HacktoberFest and GSSoC: **Rank 506** among 17,000+ participants; **20 PRs merged**, 6 badges.